Financial Dataset Curation & Prompt Creation Guidelines

1. Purpose & Objectives

The purpose of this project is to curate and generate high-quality financial dataset items that test whether an AI model can perform realistic, analyst-level financial work. Each item must represent a real professional workflow rather than a textbook exercise, simple data extraction task, or generic financial explanation.

The workflow forces the model to read multiple documents, extract relevant financial data, resolve discrepancies, apply custom financial methodologies, perform multi-step cascading calculations, and present professional, objectively gradable outputs.

To "hill-climb" and improve performance, we need to feed the model training examples that reflect the complexity and scale of the real-world evaluation set.

Core Target Failure Areas

You will be evaluating failure areas in the response that expose two primary failure modes which are:

- **Financial Methodology & Logic:** Failure to execute complex, multi-step calculations, handle methodological ambiguity, or strictly adhere to user-specified financial rules.
- **Data Extraction & Parsing:** Susceptibility to "headline bias," missing critical details in footnotes/appendices, or failing to maintain constraints across data-dense documents.

2. Domain-Specific Prompt Direction

All prompts must be built across the following four verified expert pillars. Items must map to the table below. We have also provided some use cases that can be used to create a prompt. Please do not restrict to these use cases and feel free to create your own as well.

Category	Prompt Focus	Strong Use Cases	Common Model Failure Patterns to Target

A. Investment Banking	Transaction analysis, valuation, capital raising, or strategic advisory	M&A accretion/dilutionSources and usesPurchase price allocationPro forma ownershipLeverage bridgeTrading comps valuationPrecedent transaction valuationIPO valuationStrategic alternatives memoSale vs. capital raise analysis	Using basic shares instead of diluted sharesIgnoring preferred equity, warrants, or convertiblesUsing headline EBITDA instead of adjusted EBITDAMixing LTM and forecast periodsApplying the wrong valuation multipleMissing transaction-related share count changes
B. Fixed Income Research	Debt instruments, creditworthiness, collateral performance, recovery analysis, yield analysis, or covenant compliance	Distressed recovery waterfallSenior vs. subordinated debt recovery analysisBond yield and price reconciliationExact compound IRR hurdle analysisCovenant headroom analysisCredit rating downgrade scenarioResidential mortgage portfolio reviewCommercial mortgage portfolio reviewCMBS/RMBS collateral performance analysisDebt maturity and refinancing risk assessment	Using coupon rate instead of yieldConfusing clean price and dirty priceIgnoring accrued interestApplying recovery proceeds in the wrong priority orderMisclassifying senior, secured, unsecured, and subordinated debtMissing revolver draws or letters of creditFailing to compound IRR correctlyMissing loan-level delinquency, LTV, DSCR, or collateral stratification constraints
C. Quantitative Finance	Calculation logic, statistical modeling, portfolio construction, options pricing, backtesting, or risk analytics	Portfolio optimizationCorrelation and covariance analysisFactor regressionVolatility modelingValue-at-RiskOptions pricingStrategy backtestDrawdown analysisScenario sensitivityHedge ratio calculation	Simple returns vs. log returnsDaily vs. annualized volatilityPopulation vs. sample covarianceIncorrect trading-day assumptionLook-ahead biasConstraint violationsMisapplied risk-free rateIncorrect compoundingWrong date alignment across assetsFailure to verify optimization constraints

D. Operational Finance & Assurance (FP&A)	FP&A workflows, budget control, variance analysis, forecasting, operating model updates, or pro forma corporate development schedules	Budget-to-actual variance bridgeForecast refreshHeadcount cost bridgeRevenue mix analysisGross margin walkOpex productivity analysisSegment-level planningPro forma integration forecastCost savings trackingTax or working capital planningCorporate development model under complex tax rules	Favorable vs. unfavorable sign errorsMixing absolute variance and percentage varianceBudget vs. forecast vs. actual confusionSegment totals not tying to consolidated totalsIncorrect treatment of one-time itemsMisclassification of fixed vs. variable costsMissing working capital impactsFailing to reconcile bridge components to total variance

3. Supporting Document Guardrails (PDF-Only)

To ensure the task is completely auditable, prompts must be entirely answerable from the attached files or explicit prompt assumptions alone. No CSV or spreadsheet uploads are supported; all structured data and tables must be integrated within attached PDF documents.

Acceptable PDF Materials

- Annual reports, 10-Ks, and 10-Qs.
- Earnings releases and investor presentations.
- Transaction announcements and bond prospectuses.
- Data tables formatted within PDFs, including debt schedules, mortgage collateral tables, budget/actuals tables, return history tables, option grids, factor return datasets, and cash-flow schedules.

Unacceptable Sources

- Live URLs, paywalled sources, or unstable web pages.
- Screenshots or image-only documents without extractable text.
- Wikipedia pages, generic research reports, or textbook-style educational material.
- Documents that render the answer solvable from a single, obvious summary table.

Complexity Metrics & Curation Thresholds

Human raters must align new examples with the following empirical complexity target ranges:

- **Prompt Length:** Must be between 2,000 and 4,000 characters (roughly 400 to 800 words).

- **PDF Document Length:** Extracted text must range from 20,000 to 500,000 characters per item (approximately 8 to 150 pages).
- **Number of Attachments:** Typically 2 to 3 distinct PDF files to force cross-referencing.
- **PDF Digit Density Bottleneck:** The single most data-dense PDF attached must have a digit density of 3.0% to 8.0% digits relative to total characters (representing roughly 85% narrative text and 15% embedded tables).

4. Prompt Quality Bars & Core Design Principles

Every prompt must satisfy these six fundamental criteria to ensure high-fidelity testing:

- **Realistic Context:** The prompt must mirror the explicit role and task of a working professional (e.g., an Investment Banking Analyst preparing a valuation memo for a VP). Avoid generic queries like "Calculate revenue growth" or "Summarize the company".
- **Multi-Step Analytical Flow:** The prompt must require 5 to 10 dependent steps where an early error cascades into later outputs. A standard flow involves: extracting data from Document A $\rightarrow$ extracting data from Document B $\rightarrow$ reconciling conflicts $\rightarrow$ applying custom definitions $\rightarrow$ performing intermediate calculations $\rightarrow$ building the final model output $\rightarrow$ performing a qualitative sanity check.
- **Document-Grounded:** Prompts must use specific file names (e.g., Company_2025_10K.pdf) instead of referencing live web searches or dynamic market data.
- **Static and Timeless:** Never use relative time designations like "today," "current stock price," or "latest quarter". State fixed dates and values directly (e.g., "Use the share price of $42.50 as of December 31, 2025").
- **Clearly Constrained:** All calculation assumptions, formatting rules, and rounding conventions must be explicit (e.g., "round percentages to one decimal place," "exclude operating leases from net debt," "use diluted shares").
- **Objectively Gradable:** The final deliverable must produce outputs that another finance expert can mathematically verify. Avoid purely subjective open-ended opinions.

5. Techniques for Building Financial Complexity within the prompt

To prevent the model from relying on generic pre-trained knowledge or simple keyword matching, incorporate these five curation techniques:

A. Custom & "Instruction-Conflicting" Definitions

Provide standard financial terms but explicitly attach non-standard, economically coherent rules to them.

> **Example:** *"Calculate Net Debt as total debt plus finance leases plus preferred equity, less cash and cash equivalents. Explicitly exclude operating leases."*

B. Cascading Calculations

Construct sequential calculations where mistakes early on compound heavily. For example, require the model to step through: Segment Sales $\rightarrow$ Revenue Adjustments $\rightarrow$ EBITDA Margins $\rightarrow$ Pro Forma Enterprise Value $\rightarrow$ Net Debt Adjustments $\rightarrow$ Diluted Share Count Conversion $\rightarrow$ Implied Per-Share Equity Value.

C. Footnote and Discrepancy Resolution

Hide essential variables deep within dense footnotes or secondary disclosures, or present conflicting data points that must be resolved using context.

> **Example:** *"Extract the outstanding share count to calculate diluted EPS. Note that basic outstanding shares appear on the cover page; you must locate and apply the fully diluted share count buried in the EPS footnote."*

D. Quantitative & Statistical Math

For Quantitative Finance or Credit Research tasks, provide complex data grids or historical return tables *within the PDFs* and mandate exact statistical executions rather than estimations. This includes portfolio optimization calculations, volatility modeling, drawdowns, and compound IRR hurdle rate metrics.

E. Qualitative "Sanity Check" Auditing

Conclude the prompt with a step requiring the model to evaluate whether its final calculated numbers make economic sense under specified conditions.

> **Example:** *"After calculating the terminal value, state whether a 3% perpetual growth rate requiring zero capital reinvestment is realistic under current macroeconomic conditions."*

6. End-to-End Curation & Submission Workflow

Every curated item must progress through this uniform sequence to ensure consistency and auditability:

1. **Select Taxonomy Target:** Identify the target financial category and its primary failure area.
2. **Gather PDF Source Material:** Select 2 to 3 PDF source files and verify they contain all required numbers.
3. **Validate Complexity:** Confirm the prompt length, PDF digit density, and character counts fall within target thresholds.
4. **Draft the Prompt:** Write the comprehensive prompt text stating all definitions, assumptions, exclusions, and rounding guidelines.

5. **Develop Independent Solution:** Manually solve the entire task before testing the model, logging key source pages, formulas, and intermediate values.
6. **Execute Model Baseline Test:** Run the current model exactly once using only the final prompt and final PDFs; save the complete first response in Markdown.
7. **Document Failure Pattern:** Compare the model's response against your independent solution and document the exact logic or extraction breakdown.

7. Drafting the Failure Explanation Narrative

Curators must provide a 3 to 4 line narrative explaining the failure pattern. The explanation must include the specific error type, the prompt/document trigger, the correct treatment, and the downstream calculation impact.

> **Standard Failure Narrative Template Example:** "Extraction/Footnote Error: The model is highly likely to extract basic outstanding shares from the cover page of Target_Company_10K.pdf rather than navigating to the EPS Footnote (Page 64) to locate fully diluted shares adjusted for outstanding warrants. This mistake is triggered by headline bias since basic shares appear prominently in the main balance sheet layout. A correct response must isolate the fully diluted share count, apply it to the Enterprise Value bridge, and note why basic shares were rejected, preventing a downstream overvaluation of the implied per-share equity price."

Working Principle

> **Every submission must answer this guiding question:** *"Would a seasoned financial professional recognize this prompt as a realistic, practitioner-grade task, and does it successfully expose a clear model vulnerability that can be objectively evaluated?"*